Knowing When to Abstain: Calibration and Selective Prediction for Machine-Learning Network Intrusion Detection

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***Abstract*—A machine-learning intrusion detector that emits one hard label gives no signal of when to trust it and fails silently on attacks absent from training. Proposing no new detector, we ask whether off-the-shelf ones know when they are wrong, and reach a dissociation between two failure modes. First, over-confidence is *task-dependent*: on the dated NSL-KDD benchmark the three standard detectors are badly miscalibrated under its train–test shift, and no post-hoc calibrator we try—Platt, isotonic, or temperature scaling—transfers, whereas on two modern flow corpora (CIC-IDS-2017 and 2018) the same task is well calibrated. Second, a blind spot on minority benign-mimicking families (NSL-KDD R2L, CIC Infiltration) recurs, but it is largely an *operating-point* failure, not a representational one: such families are nearly missed at the global decision threshold even though their attack score stays discriminable, and a per-family threshold recovers much of the recall—so the dramatic "detection collapse" reported on unknown families is mostly a thresholding artifact. We grade this: the effect is cleanest for the strongly scored R2L and weaker for Infiltration, whose score is itself only moderately discriminative. Selective prediction cuts selective risk at fixed coverage, but only for models whose confidence ranks errors (tree ensembles, not logistic regression); and on the benign-mimicking blind spot the lever is the operating point—a budget-tuned threshold recovers most of what the argmax default misses, while an added abstention/novelty stack gives no gain at matched cost. We also flag that the usual fixed-partition confidence intervals understate uncertainty—resampling the evaluation distribution widens them severalfold. Selective prediction is a cheap, model-agnostic safety layer whose limits we measure rather than assume.**

***Keywords*—intrusion detection, selective prediction, calibration, uncertainty, open-set recognition, NSL-KDD**

I. Introduction

A network intrusion detector that emits only "attack" or "benign" gives an operator no way to tell a confident verdict from a guess. In practice this matters twice over: alerts compete for scarce analyst attention, and genuinely novel attacks—absent from any training corpus—are exactly the cases a static classifier is least equipped to label. A detector that could say "I am not sure" on these inputs, and defer them to a human or a secondary control (Fig. 1), would be more useful than one forced to commit.

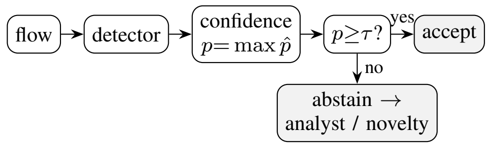

Fig. 1: The selective-prediction (reject-option) wrapper we evaluate: accept the detector's label when its confidence p clears a threshold τ, otherwise abstain and defer to an analyst or a benign-only novelty stage. We ask whether p is trustworthy—and what becomes of attacks that look benign.

This paper asks a narrow, practical question: do off-the-shelf machine-learning intrusion detectors know when they are wrong—producing confidence scores trustworthy enough to drive an abstention policy, and behaving safely on attacks they have never seen? *We propose no new detector; we evaluate existing ones.* We answer empirically on NSL-KDD [1], whose published test split deliberately contains attack types absent from training, and on two modern flow corpora (CIC-IDS-2017 and CIC-IDS-2018) under cross-day and cross-corpus drift (Section IV-E). Our finding is a *dissociation* between two failure modes, which we make precise with threshold-free analysis:

- **Over-confidence is task-dependent.** The three standard detectors are badly miscalibrated under NSL-KDD's shift, and *no* post-hoc calibrator—Platt [2], isotonic, or temperature scaling [3], all fit on source data—transfers; yet on both modern corpora the same task is well calibrated (Section IV-A).
- **The benign-mimicry blind spot is an operating-point failure, not a representational one, and it recurs across all three corpora.** Minority benign-mimicking families (R2L, Infiltration) are nearly missed at the global decision threshold, but their attack score stays discriminative (AUROC well above chance) and a per-family operating point recovers much of the recall; class balancing does not help. Threshold-free, the much-cited "detection collapse" on unknown families is mostly an artifact of the threshold (Sections IV-B, IV-D).
- **Selective prediction is a useful but bounded safety layer.** It cuts selective risk substantially, but only for models whose confidence ranks errors [4], [5]; split-conformal coverage erodes under the same shift; and on the benign-mimicking blind spot the operating point is the lever—a budget-tuned threshold recovers most of what the argmax default misses, while an added abstention/novelty stack gives no further gain at matched analyst cost (Sections IV-C, IV-G).

Our protocol and code are released anonymized and reusable (open-licensed at camera-ready). Throughout we follow the evaluation guidance of Arp *et al.* [6]: calibrators are fit only on held-out training data, and no test information leaks into model selection. Unless stated otherwise, NSL-KDD metrics are means over multiple seeds with 95% CIs; because the canonical test partition is *fixed*, those CIs reflect only train/model variance, and we report separately a bootstrap over the evaluation distribution that is roughly fivefold wider (Section IV-A).

II. Related Work

Calibration. Modern classifiers often output probabilities that do not match empirical accuracy [3], [7]. Expected calibration error (ECE) summarises this gap, and temperature/Platt scaling are common post-hoc fixes [2]. These analyses typically assume the evaluation data is drawn from the training distribution—an assumption that fails for intrusion detection.

Selective prediction. A classifier with a *reject option* [4]—it may decline to answer—trades coverage (the fraction of inputs it answers) for higher accuracy on what it does accept; the risk–coverage curve and its area [5] quantify the trade-off. The model's highest class probability (its confidence) is a strong baseline score [8].

ML for intrusion detection. The limits of learned detectors—especially on novel attacks and under distribution shift—were argued early [9] and remain a methodological hazard [6]. We do not propose a new detector; we ask whether existing ones know when they are wrong.

Unknown attacks and shift. The failure of closed-world classifiers on inputs unlike their training data motivates out-of-distribution and open-set detection, for which the maximum softmax probability is the standard baseline [8]. NSL-KDD operationalises this within a security benchmark: its test split deliberately contains attack types absent from training [1], so measuring behaviour on them needs no synthetic construction. Our family-holdout protocol complements that built-in novelty with a controlled version that makes the "unknown" condition causal. We deliberately reuse only off-the-shelf scores and standard metrics so that the conclusions concern the *evaluation* of detectors, not a new detector design.

III. Methodology

Data. NSL-KDD [1] provides 125973 training and 22544 test records with 41 features (three categorical) and a binary normal/attack label; each attack maps to one of four families (DoS, Probe, R2L, U2R). Its test split contains 17 attack *types* (3750 records) that never appear in training—a built-in open-set test (the model must face attack kinds it never saw). Numeric features are standardised and categorical features one-hot encoded (turned into 0/1 indicator columns); the encoder is fit on training data only.

Models. We use scikit-learn [10] logistic regression (a linear model), a random forest, and histogram gradient boosting (two ensembles of decision trees)—deliberately standard detectors, since our subject is their confidence, not their architecture.

Calibration. For each model we report ECE (15 equal-width bins of the predicted class probability) raw and after three post-hoc calibrators—Platt (sigmoid) scaling, isotonic regression, and temperature scaling [3]—each fitted on a 20% held-out split of the training set, both in-distribution and on the shifted test set.

Per-family detection and operating points. To separate a thresholding artifact from genuine indistinguishability, we report, besides operating-point recall, the *threshold-free* detection AUROC of the attack score (benign vs. family) per family. The binary target is near-balanced (46.5% attack), so the within-attack rarity of R2L/U2R is the only imbalance; we therefore also report recall under `class_weight="balanced"` and at a per-family threshold set to a 10% benign false-positive rate, isolating the effect of the global argmax cut. Reported separately, a bootstrap that resamples the *test partition* (B=1000) gives an evaluation-distribution CI to complement the seed CIs.

Selective prediction. The confidence of a prediction is its maximum class probability. Sorting the test set by confidence and accepting the most-confident fraction c yields the selective risk $r(c)$; we report the area under $r(c)$ (AURC) and $r(0.8)$ (Fig. 2).

Open-set protocol (LOFO). To probe genuinely unknown attacks under control, our leave-one-family-out protocol removes one attack family entirely from training (keeping benign traffic and the other families), retrains, and measures detection and confidence on that family's test records. We also report, at a global threshold giving 80% coverage, the fraction of unknown-family records that abstention rejects. Reported metrics are means over multiple seeds with 95% confidence intervals.

IV. Results

A. Over-confidence under shift, and no calibrator fixes it

All three detectors are markedly over-confident on the benchmark's shifted test set, with ECE from $\approx$0.16 (random forest) to $\approx$0.22 (logistic regression) (Table I). This is induced by distribution shift, not intrinsic: on a held-out *in-distribution* split the same models are nearly perfectly calibrated (ECE $\lesssim$ 0.01). The reviewer-relevant point is that no post-hoc calibrator fit on source data removes the shift-induced miscalibration: Platt, isotonic, and temperature scaling all leave test ECE essentially where it was (gradient boosting 0.1806 raw vs. 0.1762/0.1702/0.1816; the random forest is slightly *worse* after every one), even though each drives in-distribution ECE below 0.01 (Fig. 4). Calibrating to the source distribution does not certify probabilities once the deployed distribution moves; this is why we turn to abstention, which needs only a usable confidence *ranking*, not calibrated probabilities.

How precise are these numbers? Because the canonical KDDTest+ partition is held fixed, our seed CIs (Table I) capture only train/model variance and look reassuringly tight. They understate generalization uncertainty by construction. Resampling the *evaluation distribution* (bootstrap over the test partition) widens the accuracy interval from a seed half-

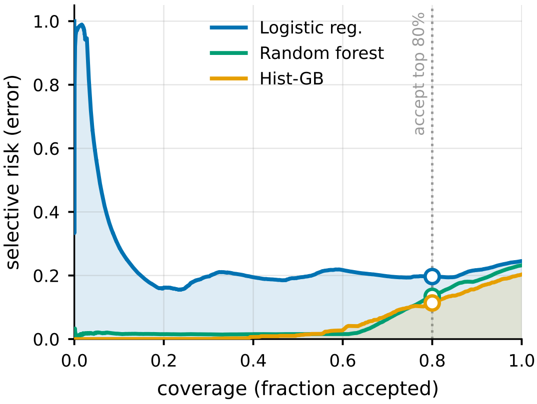

Fig. 2: Risk–coverage on NSL-KDD; the ring marks the 80%-coverage operating point and the shaded areas are the AURC (area under the curve) for the two extremes. Accepting only the most-confident inputs lowers selective risk sharply for the tree ensembles but barely for logistic regression, whose confidence does not rank its errors (nearly flat, and worst at the very top).

width near 0.001 to $\pm$0.0052—about fivefold—which is the honest scale of uncertainty for a single fixed benchmark.

B. The benign-mimicry blind spot is an operating-point failure

Our central finding is a dissociation, and it begins by dismantling the most attackable reading of the blind spot. The natural worry is that R2L's near-zero detection is just an unbalanced classifier cut at argmax (the default 0.5 decision boundary). It is not class imbalance—but it *is* largely the threshold. The attack score separates R2L from benign well above chance (AUROC 0.872 when R2L is seen in training, 0.804 when it is held out; Fig. 3), yet at the global argmax cut only 0.059 of R2L is flagged. Training with `class_weight="balanced"` barely moves this (0.064)—expected, since the binary attack/normal task is already near-balanced—whereas a per-family threshold at a 10% benign false-positive rate recovers 0.493. So R2L is the hardest family, but it is *discriminable*: the catastrophic recall is one global operating point being wrong for a minority, benign-mimicking family, not a representational dead end. This corrects an over-strong "structural blind spot" reading; the blind spot is real but it lives in the decision threshold.

The same lens deflates "detection collapse." Threshold-free, holding a family out of training costs little AUROC—DoS 0.988→0.927 (a drop of 0.061), Probe 0.01, R2L 0.068 (Table II)—even though operating-point recall falls steeply. The much-cited recall collapse on unknown families is therefore mostly the global threshold mismatching the now-unknown family, not the score losing discriminative power. Operationally this matters: the fix is a better operating point or abstention, not richer features.

C. Abstention helps—if confidence is informative

Abstaining on the least-confident inputs lowers selective risk—but only when the model's confidence ranks its errors (Fig. 2). For gradient boosting, rejecting the least-confident 20% of inputs lowers selective risk to 0.1130 $\pm$ 0.0047 (AURC 0.0530 $\pm$ 0.0027); the random forest is similar (AURC 0.0604 $\pm$ 0.0013). Logistic regression is the cautionary case: its confidence barely ranks its errors (AURC 0.2259 $\pm$ 0.0115), so abstention buys little (0.1955 $\pm$ 0.0008). The lesson is that selective prediction is only as good as the underlying confidence signal, which here depends strongly on the model family.

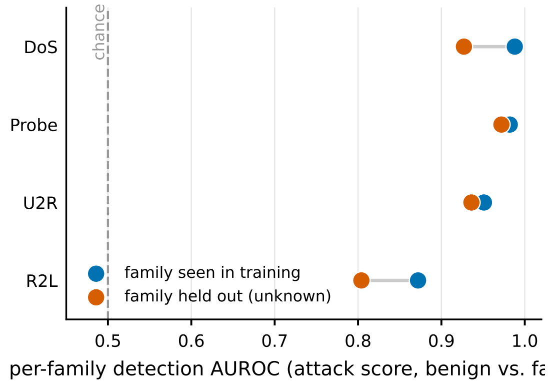

Fig. 3: Per-family detection AUROC (threshold-free), family seen vs. held out of training. Every family stays well above chance, R2L lowest but never near 0.5; the small seen→unknown gaps are the honest size of "detection collapse" once the decision threshold is taken out of the picture.

D. Leave-one-family-out (LOFO): the operating-point view

Section IV-B measured detection threshold-free; here is what an operator actually sees at a fixed operating point, under the same controlled *leave-one-family-out* (LOFO) protocol: for each family F, remove all of F from training (keeping benign and the other families), retrain, and measure operating-point detection and abstention on F's test records. At argmax, detection of DoS falls from 0.863 (seen) to 0.603 (unknown) and Probe from 0.788 to 0.426 (Table III); abstention at 80% coverage then rejects 0.406 and 0.371 of the unknown family, recovering part of the gap. R2L is the worst case—0.003 detected when unknown, only 0.168 rejected by abstention—because R2L overlaps the benign feature manifold for two independent benign-only detectors alike (Fig. 5), and confidence abstention fires on *unusual* inputs, which R2L is not. As Section IV-B established, this is an operating-point and manifold-overlap effect, not an uninformative score (R2L still scores AUROC 0.804 when unknown); the recalls here are the deployment-time consequence of that mismatch.[1] The two modern-corpus studies below repeat the protocol with the Web (CIC-2017) and Infiltration (CIC-2018) families held out.

E. Two modern corpora: calibration recovers, the blind spot recurs

The dissociation holds across two further corpora, taking the benign-mimicry result beyond n=2. **CIC-IDS-2017.** Within a random split this flow-based corpus (78 features; DDoS, PortScan, Web) is far easier than NSL-KDD—gradient boosting reaches accuracy 0.9998 at ECE 0.0001—so overconfidence simply disappears: calibration quality is a property of the task, not an intrinsic flaw (Table IV). Abstention's

[1] U2R (train n=52, test n=67) is too rare for its per-seed estimates to support qualitative claims; we list it in Table III for completeness only and draw no conclusions from it.

TABLE I: NSL-KDD test metrics: mean ± 95% CI over K=15 seeds (resampling the fit/calibration split and model randomness). Lower ECE and AURC are better.

Model	Acc.	ECE	AURC	$r(0.8)$
Logistic reg.	0.752 ± 0.001	0.2151 ± 0.0008	0.2259 ± 0.0115	0.1955 ± 0.0008
Random forest	0.769 ± 0.004	0.1630 ± 0.0036	0.0604 ± 0.0013	0.1328 ± 0.0026
Hist. grad. boost.	0.799 ± 0.004	0.1806 ± 0.0037	0.0530 ± 0.0027	0.1130 ± 0.0047

TABLE II: Threshold-free per-family detection on NSL-KDD: attack-score AUROC (benign vs. family) when the family is seen vs. held out, its change Δ, and operating-point recall at the global argmax cut. The score is discriminative for every family (AUROC $\gg$ 0.5), yet argmax recall is low for the minority families—the global cut, not the score, is what misses them. Means over K=6 seeds; AUROC CI half-width $\leq$ 0.047.

Family	AUROC (seen)	AUROC (unk.)	Δ	recall @argmax
DoS	0.988	0.927	0.061	0.864
Probe	0.982	0.972	0.01	0.781
R2L	0.872	0.804	0.068	0.077
U2R	0.951	0.936	0.015	0.177

TABLE III: Leave-one-family-out (LOFO) on NSL-KDD, *operating-point* (argmax) recall: detection when a family is seen in training vs. held out (unknown), and the fraction of unknown-family records abstention rejects at 80% coverage. These are the deployment-time consequences of the threshold-free AUROCs in Table II. Means over K=10 seeds; 95% CI half-width $\leq$ 0.064.

Family	det. (seen)	det. (unknown)	rejected by abstention
DoS	0.863	0.603	0.406
Probe	0.788	0.426	0.371
R2L	0.073	0.003	0.168
U2R	0.175	0.049	0.61

TABLE IV: CIC-IDS-2017: replication (random split) and cross-day drift (train Friday, test Thursday). Means over K=5 seeds; 95% CI half-width $\leq$ 0.0219.

Model	Acc.	ECE	AURC	$r(0.8)$
Logistic reg.	0.9831	0.0157	0.0016	0.0017
Random forest	0.9998	0.0007	0.0	0.0
Hist. grad. boost.	0.9998	0.0001	0.0	0.0
Cross-day: Web detection 0.9816→0.7916 (unknown); abstention rejects 0.9516 of unknown Web at 80% coverage.				

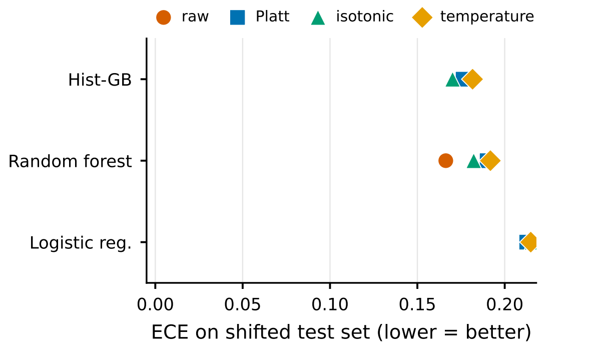

Fig. 4: No post-hoc calibrator survives the shift. Per model, ECE on the shifted test set under raw, Platt, isotonic, and temperature scaling; all four markers cluster high. Each method reaches ECE below 0.01 in-distribution, so the failure is the shift, not the calibrator.

value persists under drift: with a cross-day split (train Friday DDoS/PortScan, test Thursday with Web unknown), which avoids the optimistic bias of a random split [11], Web detection falls from 0.9816 to 0.7916 and abstention rejects 0.9516 of unknown Web—more than NSL-KDD's R2L, because Web flows, unlike R2L, do not mimic benign traffic.

CIC-IDS-2018 (third corpus). Its Infiltration day (150000 flows) is the textbook benign-mimicking family and shows the *same qualitative* pattern as R2L: well calibrated (ECE 0.006, like CIC-2017), yet the hardest family, under-detected at the global argmax cut (0.251) and only partly recovered by a per-family 10% benign-FPR threshold (0.394). We grade the dissociation rather than claim it uniform, though: Infiltration's threshold-free score is only *moderately* discriminative (AUROC 0.708), well below R2L's 0.872, and its recovered recall trails R2L's 0.493. So the operating-point reading is cleanest for R2L; for Infiltration part of the miss is *representational* —the score itself is weaker—not purely a thresholding artifact. The honest three-corpus picture is therefore a *graded* dissociation: calibration failure is task-specific, and a single global operating point under-detects minority benign-mimicking families whose score stays at least partly discriminable—most cleanly when that score is strong (R2L), less so when it is weak (Infiltration).

Cross-corpus transfer. Training on CIC-2017 and testing on CIC-2018 (the 27 CICFlowMeter features whose names match across releases) is the strongest external-validity test: the phenomena transfer but the guarantees do not. Accuracy falls to 0.674, calibration collapses (ECE 0.2625), and abstention recovers far less than within-corpus (selective risk 0.326→0.269), with Infiltration again the blind spot (detected 0.142; K=5 seeds, CI half-width $\leq$ 0.0184).

F. Which signal, and can coverage be guaranteed?

Is max-softmax (MSP) the right abstention signal, and can abstention carry a formal guarantee? On NSL-KDD, MSP is a strong, well-rounded default among classical signals: random-forest ensemble disagreement is statistically indistinguishable from it (paired ΔAURC -0.0002 ± 0.0003, CI includes 0), Mahalanobis distance ranks errors marginally better (-0.0022 ± 0.0014) but separates unknown attacks worse, and kNN is worse on both (+0.0072 ± 0.0030); the leverage is in the abstention framework, not the estimator. We compare classical signals only—energy- and ODIN-style scores assume a neural net, which we deliberately do not use, so our claim is scoped to off-the-shelf tabular models. For a guarantee, split conformal prediction [12] attains target 0.90 coverage in-distribution (0.930 ± 0.001) but only 0.611 ± 0.002 under the train–test shift [13]—the conformal analogue of the Section IV-A calibration failure; restoring it needs shift-aware weighting with estimable likelihood ratios, impractical for genuine zero-days.

G. A benign-only novelty stage partially closes the blind spot

Confidence-based abstention barely flags R2L (Section IV-D) because such attacks lie on the benign feature manifold (Fig. 5). The apt detector for that regime is a *benign-only*

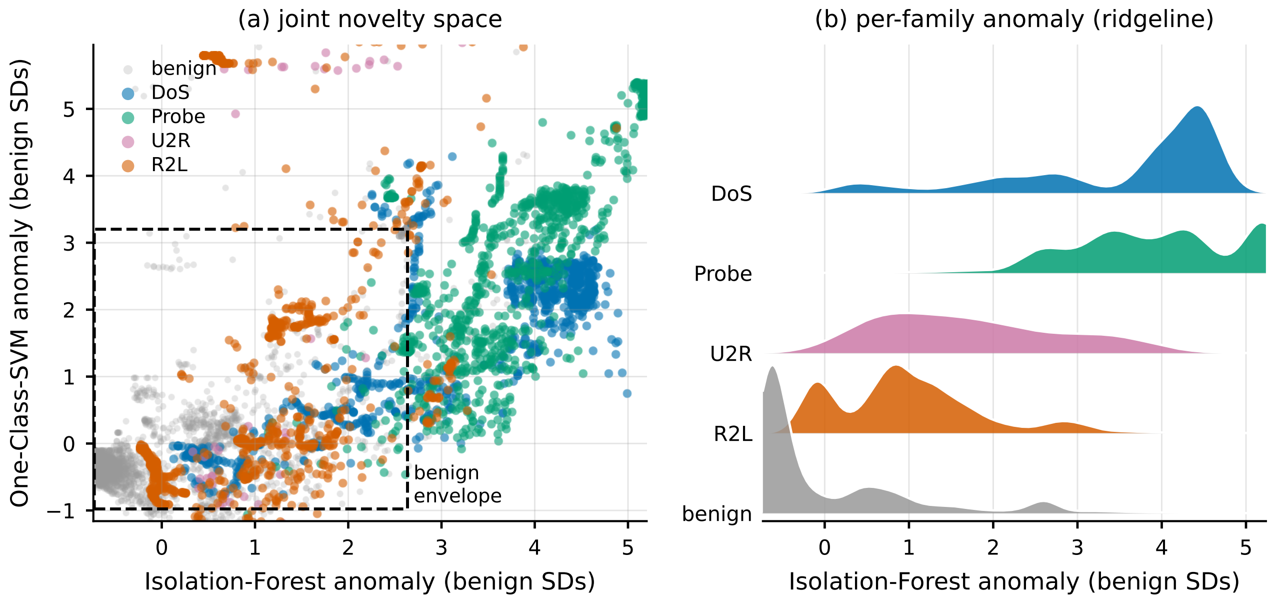

Fig. 5: Why R2L is the blind spot. **(a)** The joint novelty space: each NSL-KDD test record placed by how anomalous the two benign-only detectors find it, z-scored so benign sits near the origin (dashed box = central 95% of benign). **(b)** A ridgeline of the per-family Isolation-Forest anomaly. Both panels show R2L's distribution sitting on top of benign's—overlapping support—while DoS, Probe, and U2R separate, so neither abstention nor a benign-only novelty detector can cleanly flag the benign-mimicking R2L.

density model—a one-class novelty detector, distinct from the two-class Mahalanobis used above for error ranking. We fit an Isolation Forest [14] and a one-class SVM [15] on benign records only. At a 10% benign false-positive budget the Isolation Forest recovers 0.513 of U2R and 0.243 of R2L (the one-class SVM 0.657 and 0.41; means over $K{=}8$ seeds, 95% CI half-width $\leq$ 0.036)—against abstention's $\approx$ 0.168 for unknown R2L, a clear if partial gain. The AUROC ceilings (Fig. 7: R2L 0.824, U2R 0.897) show the limit is real but not absolute: R2L is hardest because its support most overlaps benign, so no benign-only detector fully recovers it, yet a novelty stage is a cheap, measured complement—abstention and novelty catch different failure modes (confident errors vs. benign-mimicking attacks).

The recipe, re-baselined honestly. Stacking looks like a win only against the argmax cut, a bad threshold for a minority family. We instead compare three policies on held-out (unknown) R2L at the *same* review budget (0.124): argmax, a global threshold *tuned* to that budget, and the full stack (argmax/abstention/novelty). Argmax is blind (0.003); a tuned threshold recovers 0.508; the full stack reaches only 0.397, *below* the tuned threshold at matched cost (paired gap -0.111; Fig. 6). The recovery is the operating point, not the gates; even so, the tuned threshold still misses about half of unknown R2L (the stack about 60%). We run this on NSL-KDD R2L only; the Infiltration case is left to camera-ready, a single-corpus result.

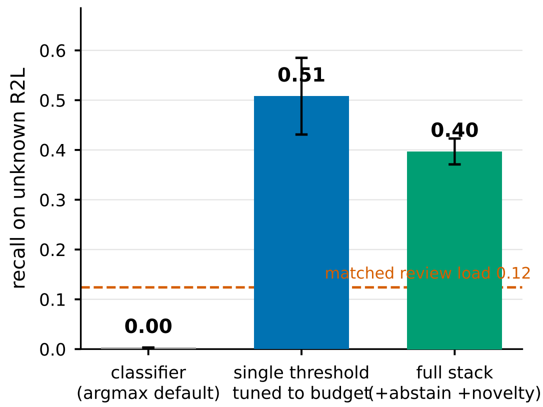

Fig. 6: The operational recipe, re-baselined. On unknown (held-out) R2L at a matched analyst review budget, the argmax default is blind (0.003); a single threshold *tuned* to that budget recovers 0.508, which the full stack (argmax + abstention + novelty) does not beat (0.397). Dashed line: the matched review load. The lever is the operating point, not the gates. Means over $K{=}6$ seeds, 95% CIs.

V. Discussion and Limitations

Selective prediction is a cheap, model-agnostic safety layer: at the cost of deferring a minority of inputs it materially lowers the error among those acted upon, and it partially flags unknown attacks. But our results bound the claim. First, the value of abstention is inherited from the confidence signal, which is good for tree ensembles and poor for the linear model. Second, post-hoc calibration on source data does not survive distribution shift, so calibrated probabilities should not be trusted as deployment-time guarantees. We should be plain about what is and is not novel here: the contribution is the *diagnosis*—that the much-cited "detection collapse" on unknown families is largely a thresholding artifact over a score that stays discriminable—not a new defense mechanism. The remedy it implies is deliberately simple (a per-family operating point plus a benign-only novelty stage), and we present it as such; the natural reaction "so just use a better threshold" is agreement with our point, not a rebuttal to it. Third, and most important operationally, a single global *argmax* operating point silently under-detects minority benign-mimicking families (R2L, Infiltration) even though their score is discriminable; the fix is to set the operating point to the analyst budget—a per-family or budget-

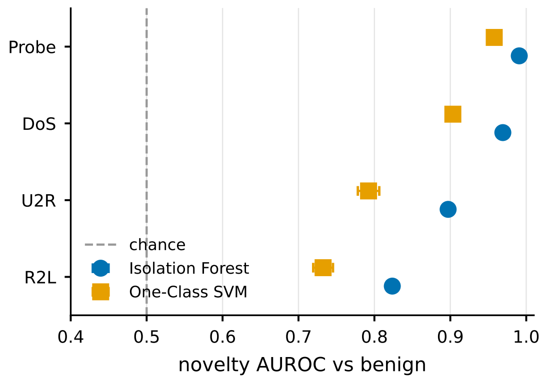

Fig. 7: Benign-only novelty detection (one-class models trained on benign records only): AUROC separating each attack family from benign, ordered by difficulty. DoS/Probe are easy; U2R/R2L—the abstention blind spot—are only partially recoverable, R2L the hardest. Markers are means over $K{=}8$ seeds, whiskers 95% CIs; the dashed line is chance.

tuned threshold, not richer features—and an added abstention/novelty stack does not beat that tuned threshold at matched cost (Section IV-G).

Operational recipe. Deploy a model whose confidence is informative (low validation AURC—a tree ensemble here, not the linear model); set the operating coverage from the risk–coverage curve to meet a target precision; route the abstained minority to a secondary control; monitor the abstention rate as a cheap, label-free drift signal [16], [17]; and, crucially, set the operating point to the analyst budget rather than leaving it at argmax—which alone recovers 0.508 of unknown R2L versus 0.003 (Section IV-G).

Limitations. NSL-KDD is dated and hand-crafted, so absolute numbers will differ on modern corpora; the mechanisms we isolate—over-confidence under shift, model-dependent confidence quality, and the operating-point blind spot on benign-mimicking families—reproduce on CIC-IDS-2017 and 2018, so they are not specific to it. We study binary detection with a single abstention signal (maximum class probability); richer signals and the multi-class attack-typing setting are natural extensions. We evaluate a cross-day split and a cross-corpus transfer (CIC-2017→2018) on shared features; a full time-ordered evaluation, the label-corrected CIC-2017 [18], and standardized NetFlow-v2 multi-network transfer (a common schema avoiding our 27-feature restriction) remain future work.

VI. Conclusion

Off-the-shelf intrusion detectors are over-confident under NSL-KDD's shift and not fixable by post-hoc calibration (Platt, isotonic, or temperature), yet well-calibrated on two modern corpora—calibration need is task-dependent. The second failure recurs on all three corpora but is an *operating-point* one: a single global threshold silently under-detects minority benign-mimicking families whose attack score is in fact discriminable, so the much-cited "detection collapse" on unknown families is mostly a thresholding artifact. A reject option lowers selective risk where confidence ranks errors, and simply tuning the operating point to the analyst budget recovers much of the benign-mimicking traffic the argmax default hides—while a stacked novelty gate adds no further recall at matched cost. Selective prediction is worth adding to ML intrusion detection, provided its confidence signal and its blind spots are measured, not assumed.

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